SIMATS ENGINEERING

Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

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Producer-Consumer Problem
10 marks

5/5 answered

Your Code

```java
import java.util.Scanner;
import java.util.InputMismatchException;
class InsufficientBalanceException extends Exception{
    public InsufficientBalanceException(String message){
        super(message);
    }
}
   class Main{
    public static void main(String[]args){
        Scanner scanner= new Scanner(System.in);
        double balance=0;
        double withdrawlAmount=0;
        boolean isValidInput=false;
        try{
            if(scanner.hasNextLine()){
                String line=scanner.nextLine().trim();
                String[]parts=line.split("\\s+");


        if(parts.length<2){
        System.out.println("Invalid input");
        }
           else{
              try{
            balance=Double.parseDouble(parts[0]);
                    withdrawlAmount=Double.parseDouble(parts[1]);
                    isValidInput=true;
                }
                catch(NumberFormatException e){
                    System.out.println("Invalid input");
                }
          }
      } else{
          System.out.println("Invalid input");
```

Input

```
10000 3000
```

Output

Online Food Delivery – Multiple Threads and Execution Order

10 marks

Q5

Manufacturing Plant – Producer-Consumer Problem

10 marks

5/5 answered

```java
import java.util.Scanner;
public class HospitalPatientRegistration{
    static class InvalidAgeException extends Exception{
    }
    static class InvalidTemperatuetrusException extends Exception{

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        try{
            String name=sc.next();
            int age=sc.nextInt();
            double temperature = sc.nextDouble();
            if(age<18){
                throw new InvalidAgeException();
            }
            System.out.println("Success");
        }
        catch(InvalidAgeException e){
            System.out.println("InvalidException");
        }catch(Exception e){
            System.out.println("invalid input");
        }finally{
            System.out.println("completed");
        }
    }
}
```

arun 35 98.6

Output

```
Success
completed
```

Submitted

5/5 answered

```java
import java.util.Scanner;
public class Main{
    public static void main(String[]args){
        Scanner sc=new Scanner(System.in);
        String[]products={"Laptop","phone","headphones"};
        double[]prices={50000,30000,2000};
        try{
            int index=sc.nextInt();
            int quantity=sc.nextInt();
            if(quantity<=0){
                throw new IllegalArgumentException("Invalid quantity");
            }
            double total=prices[index]*quantity;
            System.out.println("Product: "+products[index]);
            System.out.println("Total Cost: "+total);
        }
        catch(ArrayIndexOutOfBoundsException e){
            System.out.println("Invalid product Index");
        }
        catch(java.util.InputMismatchException e){
            System.out.println("Invalid numeric input");
        }
        catch(IllegalArgumentException e){
            System.out.println("Qunatity must be greater than 0");
        }
        System.out.println("Transaction processing completed.");
    }
}
```

Submitted

Output

```
Product: phone
Total cost: 60000.0
Transaction processing completed.
```

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JAVA PROGRAMMING

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5/5 answered

```java
import java.util.Scanner;
public class OnlineFoodDelivery{
    static boolean paymentDone=false;
    public static void main(String[] args) throws Exception{
        Scanner sc=new Scanner(System.in);
        String order=sc.next();
        String payment=sc.next();
        String delivery=sc.next();
        Thread orderThread=new Thread(()->{
            System.out.println(order);
        });
        Thread paymentThread=new Thread(()->{
            System.out.println(payment);
            paymentDone=true;
        });
        Thread deliveryThread=new Thread(()->{
            if(paymentDone){
                System.out.println(delivery);
            }else{
                System.out.println("Not proceed");
            }
        });
        orderThread.start();
        orderThread.join();
        paymentThread.start();
        paymentThread.join();
        deliveryThread.start();
        deliveryThread.join();
    }
}
```

Pizza
UPI
agents

Output

```
Pizza
UPI
agents
```

Submitted

Manufacturing Plant – Producer-Consumer Problem
10 marks

5/5 answered

Your Code

```
class Buffer{
    String item;
    synchronized void
        produce(String p) throws InterruptedException{
        while(item !=null)
            wait();
    }
        item=p;
        System.out.println("Produced: "+p);
        notify();
    }
    synchronized void consume()
        throws InterruptedException{
        while(item == null)
            wait();
    }
        System.out.println("Consumed: "+item);
        item=null;
        notify();
}
}
public class Main{
    public static void main(String[]args) throws InterruptedException{
        Buffer b=new Buffer();
        Thread producer= new Thread(()->{
            try{
                b.produce("P1");
                b.produce("P2");
                b.produce("P3");
            }catch(InterruptedException e){
                e.printStackTrace();
            }
```

Input

stdin input

Output